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03/01/26

Pratical : 8
Test based on distribution : $H_0: \sigma^2 = \sigma_0^2$ For μ unknown

Q1 The following data represent the marks obtained by 8 students in a class test:

Marks: 42, 45, 40, 44, 43, 41, 46, 44 Using MS Excel, test at the 5% level of significance whether the population variance is equal to 4.

1.) Compute the sample mean using Excel
2.) Calculate $(X_i - \bar{X})^2$
3.) Find $\Sigma(X_i - \bar{X})^2$
4.) Compute the chi-square statistic
5.) Use Excel to obtain the critical value.

Ans: i.) Sample mean ($\bar{x}$) Average() : 43.125
ii.) Calculate $(x_i - \bar{x})^2$

x_i	$(x_i - \bar{x})^2$
42	1.265625
45	3.515625
40	9.765625
44	0.765625
43	0.015625
41	4.515625
46	8.265625
44	0.765625
	28.875

iii) $\Sigma(x_i - \bar{x})^2 = 28.875$

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iv.) Compute the chi-square statistic

chi-square statistic $(n-1) * s^2 / sigma^2$

$s^2 = \Sigma(x_i - \bar{x})^2 / (n-1) = 4.125$

chi-square statistic $= (n-1) * s^2 / sigma^2) = 7.21875$

v.) critical value

chi square statistics : CHISQ.INV.RT(0,05,7) = 14.06714.

Q2 ~~A candy manufacturer claims that the distribution of colors in its assorted candy packets follows the proportions: Red: 30%, Blue: 25%, Green: 20%, Yellow: 25%. To verify this claim, a random sample of 400 candies was taken, and the observed frequencies of colors were as follows:~~

Q2 The weights (in kgs) of 7 cement bags are
Weights : 50.5, 49.8, 50.2, 49.9, 50.1, 49.7, 50.6
Using MS Excel, test at the 10% level of significance whether the population variance is greater than 0.25.

i. Compute the sample mean using Excel.
ii.) Calculate $(x_i - \bar{x})^2$
iii.) Find $\Sigma(x_i - \bar{x})^2$
iv.) Compute the chi-square statistic
v.) Use Excel to obtain the critical value.

Ans:

i.) Sample mean(x) : = AVERAGE() ⇒ 50.11429
ii.) Calculate $(x_i - \bar{x})^2$

x_i	$(x_i - x)^2$
50.5	0.14877551
49.8	0.09877551
50.2	0.007346939
49.9	0.045918367
50.1	0.000204082
49.7	0.171632653
50.6	0.235918367
Total	0.708571429

iii.) $\Sigma (x_i - x^-)^2 = 0.708571429$

iv.) Chi-square statistic : $(n-1) * s^2 / sigma^2$

$\therefore s^2 = \Sigma (x_i - x^-)^2 / (n-1)$

$= 0.118095$

$\therefore$ chi square statistics : 2.834286

v.) critical value

Chi-square statistic {CHISQ.INV.RT(0.05,7)

$\therefore$ chi-square statistic = 10.64464

Q3 The daily sales (in units) recorded for 6 days are: Sales: 120, 118, 122, 121, 119, 120. Using MS Excel, test at the 5% level of significance whether the population variance is less than 4.

i.) compute the sample mean using Excel

Ans: ii) Calculate $(x_i - x^-)^2$

iii) Find $\Sigma (x_i - x^-)^2$

iv.) Compute the chi-square statistic

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v.) Use Excel to obtain the critical value vi.) state your conclusion

i.) Sample mean (x) : = AVERAGE() $\Rightarrow$ 120

ii.) $(x_i - x^-)^2$

x_i	$(x_i - x)^2$
120	0
118	4
122	4
121	1
121	1
120	0
Total	10

iii.) $\Sigma (x_i - x^-)^2 = 10$

iv.) Chi-square statistic : $(n-1)*s^2/sigma^2$

$\therefore s^2 = \Sigma (x_i - x^-)^2/(n-1)$

$= \cancel{1.666667}\ 2.5$

chi-square statistic = 2.5

v.) Critical value :

Chi-square statistic = {CHISQ.INV (0.05, 5)}

$\therefore$ chi-square statistic = 1.1454762

vi.) Conclusion: We fail to reject H_0 (Reject H_0 if $\chi^2 < 1.145$)

Here, $2.5 > 1.1454762$

There is not enough evidence at the 5% level to conclude that the population variance is less than 4.

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Q.4 The following observations are taken from a normal population:

Observations: 15, 14, 16, 13, 15, 14, 16, 15, 14. Using MS Excel test at the 5% level of significance whether the population variance is equal to 1

1.) Compute the sample mean using Excel
2.) ~~Comu~~ Calculate $(x_i - \bar{x})^2$
3.) Find $\Sigma(x_i - \bar{x})^2$
4.) Compute the chi-square statistic
5.) Use Excel to obtain the critical value

Ans: i.) Sample mean : =AVERAGE() = 14.66667

ii.) Calculate $(x_i - \bar{x})^2$

x_i	$(x_i - \bar{x})^2$
15	0.111111
14	0.444444
16	1.777778
13	2.77778
15	0.111111
14	0.444444
16	1.777778
15	0.111111
14	0.444444
total	8

iii.) $\Sigma(x_i - \bar{x})^2 = 8$

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iv.) ~~Compute~~ Chi-Square statistic: $(n-1) * s^2 / sigma^2$

$\therefore s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n-1} = \frac{8}{8} = 1$

$\therefore$ chi-square statistic: 8

v.) Critical value:

chi-square statistic =

$\therefore$ chi-square statistic =

Q.5 The following are the lifetimes (in hours) of 10 bulbs
Life time: 101, 98, 100, 99, 102, 97, 103, 100, 98, 101
Using MS Excel, test at the 5% level of significance whether the population variance is equal to 4.

1.) Compute the sample mean using Excel
2.) Calculate $(x_i - \bar{x})^2$
3.) Find $\Sigma(x_i - \bar{x})^2$
4.) Chi-Square statistic
5.) Use Excel to obtain the critical value.

Ans: i.) Sample mean: = AVERAGE() =

ii.) Calculate $(x_i - \bar{x})^2$

x_i	$(x_i - \bar{x})^2$
101	
98	
100	
99	
102	
97	
102	
100	
98	
101	

iii.) $\Sigma(x_i - \bar{x})^2 =$

iv.) chi-square statistic : $(n-1) * s^2 / sigma^2$

$$\therefore s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n-1} =$$

$\therefore$ chi-square statistic =

v.) Critical value:

chi-square statistic =

$\therefore$ chi-square statistic =